

Mock Examination 1 (Lectures 1–4)

ISLP Chapters 1–3: Statistical Learning · K -Nearest Neighbours · Linear Regression

Duration: 90 minutes

Total credit: 90 points

Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: _____

Matriculation no.: _____

Problem	P1	P2	P3	P4	P5	Total
Maximum points	15	15	12	24	24	90
Achieved points						

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Fundamentals of statistical learning

(15 points)

Lecture slides: Chapters 1–2 — Introduction and Statistical Learning (Lectures 1–2).

- (a) [4 P] Explain the difference between using a statistical learning method for *prediction* and for *inference*. Give one concrete example of a question for each of the two goals.
- (b) [4 P] Contrast *parametric* and *non-parametric* methods for estimating the unknown function f in $Y = f(X) + \varepsilon$: describe the two approaches briefly and state one advantage of each. Which general trade-off between *flexibility* and *interpretability* lies behind this choice?
- (c) [3 P] Classify each of the following scenarios as *regression*, *classification* or *unsupervised learning*, with a one-sentence justification:
- (i) A telecom provider has monthly usage, contract length and age for 800 customers and wants to predict whether each customer will cancel the contract next month.
 - (ii) An economist records profit, number of employees and industry for 90 firms in order to understand which of these factors are associated with CEO salary, and how strongly.
 - (iii) An online retailer has the shopping baskets of 5,000 customers and wants to discover groups of customers with similar purchasing behaviour.
- (d) [4 P] Explain why the training MSE of a method is in general *smaller* than its test MSE. What is *overfitting*, and how does it show up when training and test MSE are plotted against model flexibility?

Problem 2: Bias–variance decomposition**(15 points)**

Lecture slides: Chapter 2 — Statistical Learning (Lecture 2).

For the prediction of a response at a fixed test point x_0 , three methods of increasing flexibility were analysed. A simulation study yielded the following values (all in squared units of Y); the noise variance is $\text{Var}(\varepsilon) = \sigma^2 = 1.00$:

	Method A (linear)	Method B (moderately flexible)	Method C (very flexible)
$[\text{Bias}(\hat{f}(x_0))]^2$	2.25	0.16	0.01
$\text{Var}(\hat{f}(x_0))$	0.25	1.44	3.60

- (a) **[4 P]** State the decomposition of the expected test MSE $E[(y_0 - \hat{f}(x_0))^2]$ into its three components and explain in one sentence each what the three components measure.
- (b) **[5 P]** Compute the expected test MSE at x_0 for each of the three methods. Which method should be chosen, and why?
- (c) **[3 P]** What is the smallest expected test MSE at x_0 that *any* method (however good) could achieve here? Justify your answer.
- (d) **[3 P]** Describe how squared bias, variance, training MSE and expected test MSE typically behave as the flexibility of a method increases, and explain how the numbers in the table illustrate the characteristic U-shape of the test MSE. (*Recall the expected test MSEs from part (b): 3.50 for A, 2.60 for B and 4.61 for C.*)

Problem 3: K -nearest neighbours classification**(12 points)**

Lecture slides: Chapter 2 — Statistical Learning: K -nearest neighbours (Lecture 2).

An online shop wants to predict whether a visitor makes a purchase. For seven past visitors, two standardised activity scores X_1 (pages viewed) and X_2 (minutes on site) and the outcome Y (purchase: Yes/No) were recorded:

Obs.	X_1	X_2	Y
1	4	3	Yes
2	2	2	No
3	6	2	No
4	4	5	No
5	1	1	Yes
6	7	4	Yes
7	5	6	No

A new visitor has scores $x_0 = (X_1, X_2) = (4, 2)$.

- (a) **[4 P]** Compute the Euclidean distance between x_0 and each of the seven observations (3 decimals; you may rank by squared distances). How is the new visitor classified with $K = 1$?
- (b) **[4 P]** How is the new visitor classified with $K = 3$? State also the estimated conditional probability $\widehat{\Pr}(Y = \text{Yes} \mid X = x_0)$ that the $K = 3$ classifier uses.

- (c) **[2 P]** How does the choice of K affect the bias and the variance of the KNN classifier (small K vs. large K)? What is the *training* error rate of the 1-nearest-neighbour classifier?
- (d) **[2 P]** What is the *Bayes classifier*, and why can it not be used directly in practice? In what sense is KNN related to it?

Problem 4: Simple linear regression by hand

(24 points)

Lecture slides: Chapter 3 — Linear Regression (Lectures 3–4).

A café chain investigates the effect of local radio advertising on sales. For $n = 5$ weeks it records the number of radio spots per week x and the weekly sales y (in 1,000 EUR):

Week i	1	2	3	4	5
Spots x_i	1	2	3	4	5
Sales y_i	6	7	11	13	18

You may use without proof: $\sum x_i = 15$, $\sum y_i = 55$, $\sum x_i^2 = 55$, $\sum x_i y_i = 195$. Consider the model $Y = \beta_0 + \beta_1 X + \varepsilon$.

- (a) **[6 P]** Compute the least-squares estimates $\hat{\beta}_1$ and $\hat{\beta}_0$, and write down the fitted regression line. Interpret both estimates in the context of the problem (with units).
- (b) **[4 P]** Using the fitted line from part (a), $\hat{y} = 2.000 + 3.000x$, compute the fitted values $\hat{y}_i$ and the residuals e_i for all five weeks, and verify that $\sum_{i=1}^5 e_i = 0$.
- (c) **[3 P]** Using the residuals from part (b), $(e_1, \dots, e_5) = (1, -1, 0, -1, 1)$, compute the residual sum of squares (RSS) and the residual standard error (RSE), and interpret the RSE.
- (d) **[6 P]** Using $\hat{\beta}_1 = 3.000$ from part (a), $\text{RSE} = 1.155$ from part (c) and $\sum_i (x_i - \bar{x})^2 = 10$, compute $\text{SE}(\hat{\beta}_1) = \text{RSE} / \sqrt{\sum_i (x_i - \bar{x})^2}$ and test $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$ at level $\alpha = 0.05$ using the t -statistic. Also give the 95 % confidence interval for β_1 . Use $t_{0.975, 3} = 3.182$ (97.5 % quantile of the t -distribution with 3 degrees of freedom). State your conclusion in one sentence.
- (e) **[3 P]** Compute the total sum of squares (TSS) and, using $\text{RSS} = 4.000$ from part (c), the coefficient of determination R^2 , and interpret R^2 .
- (f) **[2 P]** Using the fitted line from part (a), $\hat{y} = 2.000 + 3.000x$, predict the weekly sales for a week with $x = 6$ radio spots. Name one reason why this particular prediction should be treated with caution.

Problem 5: Multiple linear regression — interpreting Python output (24 points)

Lecture slides: Chapter 3 — Linear Regression (Lectures 3–4).

A consultancy analyses the salaries of $n = 60$ employees of a software company. The variables are **salary** (annual salary in 1,000 EUR), **experience** (professional experience in years) and the dummy variable **female** (= 1 for female employees, = 0 for male employees). The model

$$\text{salary} = \beta_0 + \beta_1 \text{experience} + \beta_2 \text{female} + \beta_3 (\text{experience} \times \text{female}) + \varepsilon$$

was fitted by least squares in Python (**statsmodels**), with the following output:

	coef	std err	t	$P > t $
Intercept	46.2000	1.100	42.000	<0.0001
experience	2.3500	0.250	9.400	<0.0001
female	-3.1000	1.550	-2.000	0.0504
experience:female	-0.8500	0.350	-2.429	0.0184

$n = 60$; $R^2 = 0.712$; adjusted $R^2 = 0.697$;

F -statistic = 46.15 on 3 and 56 degrees of freedom, $\text{Prob}(F\text{-statistic}) < 0.0001$.

- (a) **[6 P]** Interpret all four estimated coefficients precisely in the context of the problem (with units), paying particular attention to the dummy variable and the interaction term.
- (b) **[4 P]** Which coefficients are significantly different from zero at level $\alpha = 0.05$? Verify for the interaction term that the reported t -value is consistent with the reported coefficient and standard error. Should **female** be removed from the model because of its p -value? Discuss briefly.
- (c) **[4 P]** Write down the fitted regression equation separately for men and for women.
- (d) **[4 P]** Predict the annual salary of (i) a man with 10 years of experience and (ii) a woman with 10 years of experience. What is the estimated salary gap between women and men at 10 years of experience, and how can it be written in terms of $\hat{\beta}_2$ and $\hat{\beta}_3$?
- (e) **[3 P]** Interpret R^2 . State the null hypothesis of the reported F -test and the conclusion that follows from the output.
- (f) **[3 P]** Model diagnostics:
- (i) A plot of the residuals against the fitted values shows a clear U-shape (positive residuals for small and large fitted values, negative in between). What does this indicate, and name one possible remedy.
 - (ii) In an extended model, the predictor **age** has a variance inflation factor of $\text{VIF} = 12$. What does this mean, and what is the practical consequence for the regression results?

End of the examination — good luck!